

Loop Engineering — Aasan Summary (Roman Urdu + English)

Source: The AI Agent Factory — “Loop Engineering: A Crash Course” (Panaversity)

🌀 Bunyadi Idea (The Big Shift)

Pehle aap coding agent ko **turn-by-turn** chalte the: aap prompt likhte, agent kaam karta, aap check karte, phir agli instruction dete. Aap khud “heartbeat” the — matlab aap hi system ko baar baar start karte the.

Loop Engineering iska ulta hai. Ab aap ek chota **system** banate hain jo *khud* chalta hai: - Subah khud start hota hai - Dekhta hai raat mein kya hua - Faisla karta hai kya karna hai - Kaam agent ko deta hai - Result check karta hai - Sirf zaroori faislay (jo risky hain) aap tak late hain

Aapki value khatam nahi hoti — bas **shift** ho jati hai do cheezon mein: 1. **Intent** — aap clearly bata dein ke aapko kya chahiye (itna clear ke result check ho sake) 2. **Accountability** — jo bhi ship ho, uski zimmedari aap lete hain

Loop beech ka kaam karta hai. Do ends (intent + accountability) hamesha aapke paas rehte hain.

🌀 4 Layers — Prompt se Loop tak ka safar

Har layer pichli ko wrap karta hai:

1. **Prompt Engineering** — sirf woh alfaz jo aap likhte hain
2. **Context Engineering** — model ek turn mein kya kya dekhta hai
3. **Harness Engineering** — code jo model ke around hota hai (tools chalata hai, errors handle karta hai)
4. **Loop Engineering** (yeh course) — poora outer cycle: system kis cheez par kaam karta hai, kab start hota hai, kaise pata chalta hai “done” hai

Chota Loop vs Bara Loop: - **Chota loop (inner loop):** ek agent ka apna cycle — model ko context do, model tools mangta hai, tools chalao, result add karo, repeat — jab tak model khud “done” na keh de. **Problem:** model apne aap ko khud judge kar raha hota hai! Isiliye “outside stops” chahiye (test, limit, checker). - **Bara loop (outer loop):** yeh manager hai — decide karta hai kaunsa kaam dena hai, kab start karna hai, kaise grade karna hai, kal ke liye kya yaad rakhna hai. Chota loop ka ek pura run = ek “beat”.

🌀 Loop ke 6 Parts (Yaad rakhne wali cheez)

Har loop mein **5 working parts + 1 memory layer** hota hai:

#	Part	Kaam
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1	Heartbeat	Schedule ya event jo loop ko start karta hai (iske bina sirf ek run hoga, loop nahi)
2	Worktree	Isolation — do agents same file overwrite na karein
3	Skill	Project ki knowledge ek jagah likhi hui (SKILL.md), taake har run zero se start na ho
4	Subagent (Maker-Checker)	Ek agent kaam karta hai, dusra check karta hai
5	Connector (MCP)	Loop asli tools mein <i>act</i> kar sake (PR khole, ticket update kare) — sirf suggest na kare
6	Spine (State/Memory)	Disk par file jo yaad rakhti hai kya hua — “ No spine, no loop ”

Yaad rahe: model har run ke baad sab bhool jata hai. Repo (files) yaad rakhti hai.

♥ Part 2 — 4 Heartbeats (Loop kaise start hota hai)

Chaar tarah ke heartbeat hain — “aap pakde huye ho” se “khud chalta hai” tak:

1. **In-session loop** (/loop) — kitchen timer jaisa. Sirf tab tak chalta hai jab tak aapka session/terminal khula hai. Session band = loop band.
 - *Example:* ISS (Space Station) ki location har minute check karna.
2. **Conditional loop / Run-until-done** (/goal) — “khana taste karke bataye ke ready hai ya nahi.” Timer se nahi, ek **checked condition** se rukta hai. Jaise: “jab tak tests pass na ho jayein.”
 - Zaroori: agent khud apna kaam approve nahi karta — ek **alag checker** (chota model) decide karta hai “done” hai ya nahi.
 - Har loop ko 3 stops chahiye: **success condition**, **limit** (max tries), **no-progress check**.
3. **Scheduled (Routine)** — alarm clock jaisa. Laptop band ho tab bhi chalta hai (Anthropic ke servers par). Jaise: “har weekday 9am, overnight CI failures dekho.”
4. **Event-driven** — doorbell jaisa. Kuch nahi hota jab tak koi PR khole ya message aaye — phir turant react karta hai.

Choose karne ka rule: Task khatam hota hai aur command prove kar sakta hai → **conditional**. Task repeat hota hai → **schedule/event**. Task sirf ek dafa hota hai → **koi loop nahi chahiye**, normal session use karo!

🔧 Part 3 — Loop Kya Karta Hai (The Body)

Yeh 4 cheezein har “beat” mein hoti hain:

- **Isolation (Worktrees):** Parallel agents ek dusre ke kaam ko overwrite na karein — har ek ko apna folder/branch milta hai.
- **Knowledge (Skills):** Project ki habits ek SKILL.md mein likhi hoti hain, taake har run se dobara explain na karna pare.
- **Action (Connectors/MCP):** Loop sirf baat nahi karta — real kaam karta hai (PR khole, Slack post kare). **Rule:** Kam tools rakho (jo zaroori nahi unhe hatao), writes safe-to-repeat hon, error messages clear hon.
- **Maker-Checker (Subagents):** Sab se important choice! **Jo agent kaam banata hai, wahi khud grade nahi karta.** Ek dusra agent (alag model bhi ho sakta hai) check karta hai — isko **LLM-as-judge** bhi kehte hain.

Verification Skills: Jo cheez aap baar baar manually check karte hain, usko ek skill mein likh dein. Yeh check 4 jagah “reh” sakta hai: 1. **Standalone** — aap khud invoke karo 2. **Embedded** — us skill ke end mein add ho jo kaam banati hai 3. **Chained** — ek skill dusri ko call kare 4. **Har PR par** — team-wide gate ban jata hai

✍ Part 4 — Spine (Memory Between Runs)

Sab se zaroori concept jo beginners skip karte hain!

Model har run ke baad **sab kuch bhool jata hai**. Agar har beat zero se start ho, to loop nahi — bas wahi pehla step baar baar chalta hai.

Do files rakho: - **Rules file** (CLAUDE.md / AGENTS.md) — hamesha ki habits, har run start mein padhi jati hai (short rakhna, kyunki har run par cost hoti hai) - **Progress file** (progress.md) — jo hua uska record: kya done hua, kya baaki hai. Har run **shuru mein padhta hai, end mein update karta hai**.

Intern ki diary wali example: Naye intern ko diary do. Front mein — sabak jo mila (mistakes se seekha), roz padho. Back mein — kal kya kiya, kahan chhoda, roz likho. Bina diary ke intern (aur loop) roz wahi galtiyan repeat karega.

🔗 Part 5 — Ek Poora Loop (Real Example)

Morning-triage loop: Har weekday 9am — overnight CI failures dekhta hai, safe fixes draft karta hai, ek reviewer check karta hai, PASS pe PR khol deta hai, risky cheez insaan ke liye chhod deta hai.

Minimum Safe Loop Checklist (7 cheezein): 1. Success condition 2. Limit (max tries/time/spend) 3. Isolated branch/worktree 4. Read-only checker 5. State file (spine) 6. Human gate (risky kaam insaan tak jata hai) 7. Log/notification (silent failure na ho)

Result: Aap subah uthte hain — 2 PRs review ke liye, 1 flagged decision. Aapne kuch type nahi kiya!

👤 Part 6 — Insaani Control Kaise Rakhein (Sabse Important Part!)

3 Feedback Loops (Ek Dusre Ke Andar)

1. **Coding loop** (minutes) — agent likhta hai, test karta hai, fix karta hai
2. **Feedback loop** (hours) — aap try karte hain, decide karte hain kya change karna hai
3. **Outside loop** (days) — real users use karte hain

Agent teenon khud nahi chala sakta — kyunki **aapko woh cheezein pata hain jo agent ko nahi** (Andrew Ng isko “context advantage” kehte hain). Machine fast loop chalata hai; aap decide karte hain **kya banana hai aur kaun zimmedar hai**.

Token Cost — Asli Limit

- Har loop ko **cap karo** (max tries/time/spend)
- Sahi model choose karo (mushkil kaam ke liye strong, aasan ke liye cheap)
- Loop ko kam frequency par chalao — har 5 min ki jagah har ghante, cost mein bohat farq aata hai!
- *Example:* Har 5 minute chalne wala loop, har ghante chalne wale se 100+ guna mehnga ho sakta hai — kaam wahi hai!

Human In / On / Out of the Loop

Term	Matlab
Human IN the loop	Har action se pehle insaan approve karta hai — slow, zyada control
Human ON the loop	System khud chalta hai, insaan dekhta hai aur rok sakta hai — fast
Human OUT of the loop	Koi dekh hi nahi raha — yeh kabhi acceptable nahi

Rule: Jahan galat move mehngi aur wapis lena mushkil ho, wahan insaan rakho. Baqi jagah loop chalne do.

Concept 15: Apne Project Ki Samajh Mat Khona!

Do log same loop bana sakte hain — ek use kar ke aur tez ho jata hai (kyunki gehri samajh rakhta hai), dusra use kar ke samajhna hi chhod deta hai. **Loop yeh farq nahi bata sakta — sirf aap bata sakte hain.**

“AI Gravity” — yeh force hamesha khinchta hai ke AI se zyada se zyada kaam karwao. Loop se yeh pull aur strong ho jata hai kyunki woh so’te waqt bhi chalta hai.

🐕 Dogfooding — Yeh Book Khud Apne Loops Use Karti Hai

1. **Feedback loop** — readers ke comments automatically sort karta hai, sirf zaroori cheezein insaan tak jati hain
2. **What’s New loop** — book mein daily changes ka summary khud likhta hai, koi human approval nahi (kyunki galti sasti hai, revert kar sakte hain)

Rule jo yahan se seekha: Jahan galti mehngi ho, wahan human gate rakho. Baqi jagah chhod do.

★ Sabse Zaroori Baat (Yaad Rakhne Wali)

Do layers hain is course mein — ek yaad rakhni hai, doosri sirf lookup karni hai: - **Lasting layer (yaad rakho):** Loop ki shape (heartbeat, 5 working parts, spine), maker-checker split, aur intent+accountability ka concept. - **Mechanical layer (lookup karo):** Commands, flags, model names — yeh har hafte change hote hain, docs check karte rehna.

Quick Glossary

Term	Aasan Matlab
Agent	AI system jo tools use kar sakta hai, sirf jawab nahi deta
Loop	System jo kaam start karta hai, check karta hai, yaad rakhta hai, repeat karta hai
Beat	Loop ka ek pura run
Heartbeat	Woh schedule/event jo beat start karta hai
Stopping condition	Testable rule jo bataye kaam done hai
Maker-checker	Ek banata hai, dusra check karta hai
Worktree	Alag working folder/branch
Spine	Saved state jo ek beat se dusre beat tak memory le jati hai
Human gate	Woh point jahan insaan approve karta hai risky kaam se pehle
Routine	Claude Code ka cloud automation

Yeh summary poore chapter (6 Parts + Dogfooding + 8 Practice Projects + Routines Appendix) ka overview hai. Agar kisi ek part ko deeply teach karna ho students ko, bata dena — us par alag se detail nikaal doonga.